Sales Forecasting & Demand Intelligence — Executive Summary

Prepared for: Head of Supply Chain & CFO | Data: Superstore Sales, 2015–2018

Executive Summary

We built a forecasting system on four years of company sales data to answer one question: how much will we sell next month, and where should we adjust stock? The short answer: the business is growing steadily, sales are highly seasonal with a strong November-December peak, and our best model predicts continued growth into the next quarter, led by the East region and Technology products. A small number of unusual sales weeks were flagged for review, and we found four distinct demand patterns across product types that call for different stocking approaches.

Key Findings

- Technology is the top revenue category (~$827K over 4 years), but all three categories — Technology, Furniture, Office Supplies — contribute fairly evenly; the business isn't dependent on one product line.
- The East region shows the most reliable growth: sales rose every single year from 2015 to 2018 without a dip. West has higher totals but a bumpier history.
- Sales are strongly seasonal — November and December are consistently the two strongest months every year, while February is consistently the weakest. This pattern should drive staffing, promotions, and inventory timing decisions.
- Order-to-ship time is consistent across regions (roughly 4 days everywhere), so shipping delay is not a regional risk factor worth separate action.

3-Month Sales Forecast

We tested three different forecasting approaches side by side and picked a winner based on accuracy against real held-out data, not preference.

Model	Avg. Error (MAPE)	Recommended?
SARIMA (statistical)	17.7%	Yes — most accurate
XGBoost (machine learning)	18.0%	Close second
Facebook Prophet	21.9%	Least accurate here

Using the recommended model, projected total company sales for the next three months are approximately:

- Month 1: ~$74,000
- Month 2: ~$95,000
- Month 3: ~$107,000

Note: these are the model's most likely estimates; actual results will fall in a range around these numbers, and a 3-month test window is a small sample, so we recommend treating this as directional rather than exact.

Where the Growth Is Concentrated

Comparing the forecast against the same months one year earlier (the fair, season-adjusted comparison), the strongest projected growth is in the East region and the Technology category. Furniture and West region growth is more modest but still positive.

Top Anomalies Detected

We ran two independent anomaly-detection methods on weekly sales. Where they agree, confidence in the finding is highest:

- Late November / early December spikes (2016, 2018) — consistent with Black Friday and holiday promotions.
- A sharp one-off spike in March 2015 that does not repeat in later years — likely a single large bulk order rather than a recurring pattern.
- Several unusually low weeks in January/February — consistent with the expected post-holiday demand drop, not a cause for concern.

Product Demand Segments & Stocking Strategy

Segment	Examples	Recommended Stocking Approach
High Volume, Stable	Chairs, Phones, Tables, Storage	Maintain steady safety stock; standard reorder rules
Growing, High Value	Copiers	Increase stock ahead of demand; monitor closely
Low Volume, Steady	Paper, Labels, Fasteners, Art	Smaller, more frequent orders to reduce holding cost
Declining	Machines	Reduce commitment; consider clearance to move existing stock

Business Recommendations

1. Build inventory ahead of the November-December peak specifically for the East region and Technology products, where growth is concentrated.
2. Reduce future stock commitment to the Machines sub-category given its sustained sales decline, and consider a clearance push on existing inventory.
3. Set up an automated weekly anomaly check (using the methods in this report) so unusual sales weeks are reviewed within days, not discovered a quarter later.

Limitation to Be Aware Of

This forecast was validated on only three held-out months due to the size of the available data. That's enough to compare models fairly, but not enough to fully guarantee accuracy quarters into the future. We recommend re-validating the model each quarter as new actual sales data comes in, rather than treating this forecast as fixed for the year.